
RESEARCH STATEMENT

My research journey began the summer I joined the University of Edinburgh as a research assistant. That first encounter with academic research sparked a fascination with the gap between machine learning theory and the practical challenges of building machine learning systems in the real world. I found myself drawn to a central question: how can we build systems that make machine learning efficient and scalable across wireless and networked environments? This question has since led me down an interdisciplinary path, from distributed systems to wireless networking, and continues to shape my research direction today.

The Beginning: Learning Systems at Scale

When I first joined the research group at Edinburgh, I was drawn to the challenge of evaluation. Distributed machine learning systems are notoriously difficult to test at scale—running experiments across hundreds of edge devices is expensive, time-consuming, and often impractical. This led me to work on emulation platforms that could simulate large-scale ML workloads without requiring physical hardware.

My initial work on EmuRAN taught me a valuable lesson about scale. Building a system capable of emulating thousands of mobile devices taught me that the gap between a working prototype and a scalable system is often underestimated. The problems that emerge at scale—resource management, network virtualization, coordination overhead—are qualitatively different from those at small scale.

The Intersection of Networking and ML Systems

Through my research, I became increasingly interested in an emerging opportunity: the growing convergence between wireless communication infrastructure and AI systems. Mobile networks already contain substantial compute resources that remain idle much of the time. What if we could harness these spare cycles for machine learning?

This question led to my work on Weaver, a system exploring foundation model training over RAN infrastructure. The core challenge was not merely technical but also fundamental—how do you train machine learning models on infrastructure that must prioritize its primary function? The work required rethinking resource allocation and scheduling to ensure that AI workloads could coexist with network operations without degradation.

I also contributed to Morphling, an emulator for distributed ML systems. Here, the focus was on building evaluation infrastructure that could keep pace with the rapid evolution of ML systems at the edge. The key insight was that emulation fidelity matters: to be useful, an emulator must faithfully represent the characteristics of real networked environments, including resource heterogeneity and communication constraints.

Looking Forward

My goal is to pursue a PhD conducting cutting-edge, high-quality research at the intersection of networked systems and AI. I am excited by the challenges that emerge when machine learning meets real-world infrastructure—from wireless networks to large-scale emulation—and I want to continue pushing the boundaries of what these systems can achieve.

I am broadly interested in AI-RAN, networked digital twins, and the systems infrastructure that makes next-generation networked intelligence possible. I look forward to developing these interests further through rigorous research and collaboration in a PhD program.